

Homework 1

Note: Submit your work (upload the .java source code files ONLY, not the compiled .class files!) through the “Homework 1” link on Microsoft Teams/Codegrade (to be clarified). You may submit an unlimited number of times; we will only grade the last/latest submission attempt, but be sure to **attach all of your files to each submission attempt**. Be sure to include your name and ID number in a comment at the beginning of each file that you submit.

Due: Friday, October 31, 11:59pm **Total: 40 points (5 points per problem)**

Submission Instructions: Name your java classes for this assignment as:

Problem1: Schedule.java,

Problem3: CarFuelConsumption.java,

Problem5: Eggs.java,

Problem7: Divisible.java,

Problem2: ShowMeTheMoney.java,

Problem4: IsLeapYear.java,

Problem6: WindChill.java,

Problem8: Triangle.java

1. (Schedule.java) Write a Java program Schedule that prints your name, ID number and weekly class schedule. It should have the general appearance of the example below with your name on the first line, followed by your ID # on the second line, followed by your weekly schedule. Adjust the time slots as necessary to correctly reflect your actual class schedule. If you are taking fewer than four courses, add fictional courses to bring the total up to four.

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Time	Monday	Tuesday	Wednesday	Thursday	Friday
8:30am-9:50am		EGL101		EGL101	
10:00am-11:20am	CSE114		CSE114		
11:30am-12:50pm					
1:00pm-2:20pm		MAT125		MAT125	
2:30pm-3:23pm	FRN111		FRN111		FRN111
4:00pm-5:20pm					
5:30pm-6:50pm		PHY127		PHY127	

2. (ShowMeTheMoney.java) Before 1961, people living in Great Britain had to deal with a pretty complicated currency consisting of pounds sterling, shillings, pence, farthings, and other coins. The currency system worked like this:
 - 1 pound sterling = 20 shillings
 - 1 shilling = 12 pence (“pence” is the plural of “penny”)
 - 1 penny = 4 farthings

Write a program (ShowMeTheMoney.java) that reads in a single integer, the number of farthings, and then prints a sequence of integers that represents the fewest number of each bill or coin needed to represent that amount, starting with the largest denomination. (Hint: use the % operator.)

For example, if the argument is 2563 farthings, then the program prints 2, 13, 4, 3 because 2,563 farthings is equivalent to 2 pounds, 13 shillings, 4 pence and 3 farthings.

Examples:

(You don't have to print the currency each number is associated with, just printing the number is fine.)

Enter in the number of farthings: 2563

Sterling 2

Shilling 13

Penny 4

Farthings 3

Enter in the number of farthings: 4900

Sterling 5

Shilling 2

Penny 1

Farthings 0

Enter in the number of farthings: 762

Sterling 0

Shilling 15

Penny 10

Farthings 2

3. (CarFuelConsumption.java) Write a program (CarFuelConsumption.java) that reads the following arguments, in this order:

- minutes: the number of minutes a car is driven on a long track
 - mph: the constant speed of the car, given in miles per hour
 - mpg: the fuel efficiency of the car, given in miles per gallon
- The function computes and prints the number of gallons of gasoline consumed by a car driven down a long track over the given time period. Use floating-point division only.

Examples

Input the number of minutes a car is driven on a long track: 75

Input the constant speed of the car in miles per hour: 35.2

Input the fuel efficiency of the car in miles per gallon: 20.6

2.1359223300970873

Input the number of minutes a car is driven on a long track: 60

Input the constant speed of the car in miles per hour: 60

Input the fuel efficiency of the car in miles per gallon: 30

2.0

Input the number of minutes a car is driven on a long track: 164.3

Input the constant speed of the car in miles per hour: 45.2

Input the fuel efficiency of the car in miles per gallon: 19.41

6.376747381075049

Hint: minutes/60 * mph * 1/mpg

4. (IsLeapYear.java) A leap year is a calendar year which contains an additional day in the month of February. The purpose of the additional day every four years is to keep our calendar in sync with the Earth's revolution around the sun. A leap year can be determined by the following conditions:

If the year is divisible by 400 then it is a leap year

If the year is not divisible by 400 but divisible by 100 then it is not a leap year

If the year is not divisible by 400, not divisible by 100 but divisible by 4 then it is a leap year

If it fails all the condition above then the year is not a leap year

Write a program (IsLeapYear.java) which prompts the user to enter in a year and determine whether or not it is a leap year or not.

Examples

Enter in a year: 2021

2021 is not a leap year

Enter in a year: 2020

2020 is a leap year

Enter in a year: 1400

1400 is not a leap year

Enter in a year: 1900

1900 is not a leap year

5. (Eggs.java) Stony Brook Dairy Farm sells organic brown eggs to local customers. It charges \$3.25 for a dozen eggs, or 45 cents for individual eggs that are not part of a dozen. Write a program that prompts a user for the number of eggs in the order and then display the amount owed with a full explanation. For example, typical output might be, ***You ordered 27 eggs. That's 2 dozen at \$3.25 per dozen and 3 loose eggs at 45 cents for a total of \$7.85.*** Save the program as Eggs.java.
6. (WindChill.java) How cold is it outside? The temperature alone is not enough to provide the answer. Other factors including wind speed, relative humidity, and sunshine play important roles in determining coldness outside. In 2001, the National Weather Services (NWS) implemented the new wind-chill temperature to measure the coldness using the temperature and wind speed. The formula is

$$t_{wc} = 35.74 + 0.621t_a - 35.75v^{0.16} + 0.4275t_a v^{0.16}$$

Where t_a is the outside temperature is measured in degree Fahrenheit, v is the speed measured in miles per hour, and t_{wc} is the wind-chill temperature. The formula cannot be used for wind speed below 2 mph or temperature below -58 Fahrenheit or above 41 Fahrenheit.

Write a program that prompts the user to enter a temperature between -58 Fahrenheit and 41 Fahrenheit and a wind speed greater than or equal to 2 then displays the wind-chill temperature. Use `Math.pow(a, b)` to compute $v^{0.16}$.

7. (Divisible.java) Write a program that prompts the user to enter an integer and determine whether it is divisible by 5 and 6, whether it is divisible by 5 or 6, and whether it is divisible by 5 or 6, but not both. Here is a sample run of the program:

```
Enter an integer: 10
Is 10 divisible by 5 and 6? false
Is 10 divisible by 5 or 6? true
Is 10 divisible by 5 or 6, but not both? true
```

8. (Triangle.java) Write a program that reads three edges (sides) for a triangle and classify it as Equilateral, Scalene, Isosceles if the input is valid. The input is valid if the sum of every pair of two edges is greater than the remaining edge. You may assume that the user inputs a positive value for the edges.

Here is a sample run.

```
Side1:3
Side2:3
Side3:9
Output: Invalid input!
```

```
Side1:3
Side2:3
Side3:3
Output: Equilateral
```

```
Side1:10
Side2:10
Side3:1
```

Output: Isosceles

```
Side1:3
Side2:4
Side3:5
```

Output: Scalene

[**Hint:** Equilateral: All sides equal, Isosceles: Two sides equal, Scalene: All sides different]